Experiment : Code I
Tension Test: Mechanical Behavior of Mild Steel

Objective of the experiment:

To find the mechanical properties of mild steel.

Introduction:

Steel, an iron-carbon alloy, is extensively used for a wide variety of engineering applications. In design of steel components or structures it is important to quantify its mechanical properties. In a simple tension test, typically a round tensile bar is subject to tensile deformation and the force-deflection curve till failure is obtained. Using the experimental data several mechanical properties of the material can be determined.

Background information:

Engineering stress in a tensile bar (S): Measure of the intensity of the internal force (Stress). It is the axial force per unit original area

$$S = \frac{F}{A_o}$$

Engineering strain (e): Measure of deformation, given by change in length per unit original length

$$e = \frac{\Delta l}{l_o}$$

True stress (σ): Axial force per unit current area

$$\sigma = \frac{F}{A} = S(1+e)$$

True strain (ε): calculated using the instantaneous gauge length

$$\varepsilon = \int_0^l \frac{dl}{l} = \ln\left(\frac{l}{l_o}\right) = \ln(1+e)$$

Experimental Procedure:

1) Find the dimensions of the round tensile bar (length of the uniform CS, average diameter).
2) Get familiarized with the machine and the devices for measurement of force and deflection. Install the specimen in the machine avoiding any significant load developing during the installation.
3) While one end is fixed, provide displacement at the other end obtained by hydraulic pressure.

4) Observe the plastic deformation, necking and finally fracture during the process of loading.
5) Record the force and the corresponding deflection. Besides these data points also record the points where any sudden change in data pattern occurs.
6) In the data table note at what value of load you first notice the localization of deformation (formation of a necked region).
7) After failure, record the characteristic features of the failure surface.

Observations:

1) Pre-test data
    a. Length of uniform cross section, L_0 :
    b. Diameters at three different location, D_1 , D_2 , D_3 :
    c. Average diameter, $D = (D_1 + D_2 + D_3) / 3$:
2) Test data

Sl No.	Applied Extension, mm	Axial Force, N	Engg stress S (N/mm^2)	Engg strain e (mm/mm)	True Stress σ (N/mm^2)	True strain ε (mm/mm)

3) Post-failure data
    a. Failure load :
    b. Final diameter :
    c. Final length of the initially uniform CS :
    d. Failure surface profile :

Report:

1) Pre-test data
2) Test data
3) Post-test data
4) Plot engineering stress (y-axis)-engineering strain (x-axis) curve as well as the true stress-true strain curve in the same graph
5) Determine
    a. Proportional limit σ_{pl}, the engineering stress when the linear portion of the stress-strain curve ends.
    b. Tangent modulus E_t, the slope of the stress-strain curve.
    c. Yield strength of the material σ_y, engineering stress that corresponds to an offset of 0.002 strain (i.e., 0.2% strain) in the stress strain curve

d. Ultimate tensile strength σ_{UTS}, highest engineering stress reached at any time during the test.
e. Percentage elongation, , measure of ductility of the material:

$$= \left(\frac{L_f - L_o}{L_o} \right) \times 100$$

f. Percentage area reduction

$$= \left(\frac{A_o - A_f}{A_o} \right) \times 100$$

6) Discuss:
   a. Why is steel a popular structural material?

   b. If the experiment was in load control (i.e., the force was increased incrementally rather than the displacement), how would the stress-strain curve look like in comparison to what you have obtained. Show schematically in a diagram.

   c. Is the tangent modulus that you obtain near to the quoted values of steel (210 GPa)? If not what you think may be the reason for it.

   d. Up to what values of strain there is less than 5% difference between true stress and engineering stress data?